

Author Declarations

Manuscript: Deep Learning Architectures for Peptide-MHC Class I Binding Prediction **Journal:** *Briefings in Bioinformatics*, Oxford University Press **Date:** June 15, 2026

Conflict of Interest

The authors declare no conflicts of interest.

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Author Contributions

Zhuha Zhou, MD (†,*) — Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing — Original Draft, Visualization

Yongyu Bai, MD (†) — Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing — Review & Editing, Validation

Qigang Xue, MD — Resources, Validation, Writing — Review & Editing

Zhuxian Zhou, PhD — Supervision, Writing — Review & Editing

Shaoliang Han, MD, PhD — Conceptualization, Supervision, Project Administration, Writing — Review & Editing, Funding Acquisition

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Data Availability

All data are included in the supplementary data package (`manuscript_data_package.tar.gz`) and available from the corresponding author upon reasonable request. Public databases used: IEDB (<https://www.iedb.org/>), MHCflurry 2.2.0 (<https://github.com/openvax/mhcflurry>).

Code Availability

All R and Python scripts are included in the supplementary data package. Key scripts:

`peptide_mhc_binding_study.R` (1,700+ lines), `label_peptides_mhcflurry.py` , `benchmark_iedb_epitopes.R` , `scan_protein_epitopes.R` , `mutation_scan.R` . Software: R 4.6.0, Python 3.10.20, TensorFlow 2.17.0, Keras 2.16.1 (R) / 3.12.2 (Python), MHCflurry 2.2.0.

Ethics Statement

Not applicable — purely computational study using publicly available datasets.

AI Usage Disclosure

Claude Code (Anthropic, June 2026) was used as a programming assistant. AI-assisted search tools supplemented manual literature review. AI assistance was used for language editing and formatting. Scientific content was authored

and approved by the human authors. No AI tools were listed as authors.